Application of machine learning to user experience : How to interpret quantitative UX questionnaires results ?

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While user experience is essential in web applications, it is still frequently underestimated

This study aims to improve the visual design of web application interfaces by identifying specific areas for improvement through the use of machine learning techniques to analyze quantitative test results.

User experience (UX) is a key aspect that must be carefully analyzed during the development of a system, as it is essential for attracting and motivating users to engage with it. There are various ways to assess the usability of an interface where testers grade its overall quality. However, techniques like surveys produce quantitative results that require careful analysis to find improvement areas, which can take time. This study aims to address this challenge by exploring methods to assist UX developers in efficiently completing this process.

Researchers have explored several approaches to improve specific aspects of user interfaces. A prominent area of research addressing this challenge focuses on developing questionnaires, each with unique structures to assess interfaces from different perspectives. The System Usability Scale (SUS) created by Brooke[1] asks testers to rate interface statements, and combines the grades into an overall SUS-score. Similarly, the User Experience Questionnaire (UEQ) by Laugwitz et al. [4] evaluates a wider range of factors like originality, complexity, and aesthetics. Researchers have also applied machine learning to this field, such as Rim et al. [5], who used Bayesian networks to analyze users preferences based on their interaction with the system, and Esposito et al. [2], who employed Random Forests and Decision Trees to predict users emotions from their interactions. These studies highlight the importance of improving interface design and demonstrate multiple approaches to achieve this goal. However, most

of them primarily focus on the direct user behavior with the system, rather than leveraging questionnaires results, which could provide more accurate and detailed data on the tested interface.

The main goal of this study is to find a technique that can interpret the quantitative data derived from testers responses to questionnaires assessing interface usability. By effectively interpreting these results, UX developers would be able to get a detailed analysis of the system elements requiring modification. A very limited number of studies have pursued this goal. One notable example is the work of Hinderks et al. [3], where the team created a graph-based technique to intuitively represent User Experience Questionnaire results. It helped to better understand whether broad aspects of the interface have been sufficiently worked on or not, compared to the goals of the overall system. While useful, this approach lacks focus on the finer details of individual interface components. Although machine learning is already applied in user experience, it has not been used to interpret questionnaire results. This study addresses this gap by testing classification machine learning models on UX questionnaires data to identify components for improvement and suggest optimization techniques.

The proposed solution is as follows : based on the feedback provided by testers after evaluating multiple interfaces, the responses are analyzed to derive a set of conclusions that will form the categories for the machine learning classification model. For example, responses from testers regarding a specific interface might lead to conclusions such as 'Intuitiveness' or 'Aesthetics', meaning that these areas should be improved in the interface. The model will then be trained and evaluated using those data, with the conclusions categories serving as predicted answers.

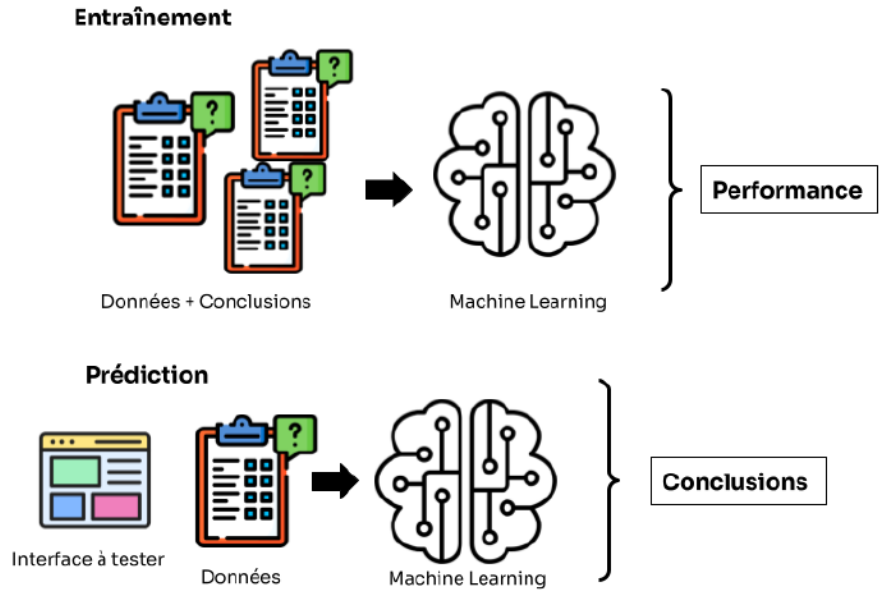


Figure 1: Schema of the proposed solution

We developed the artefact based on the Attrakdiff questionnaire due to its level of detail. A set of answers was obtained from a human-computer interaction class, where several students tested different interfaces using it. We processed and cleaned the collected data, removing empty values and standardizing them to ensure compatibility with the machine learning models. We tested in total six models to find the most optimal one, each using two different training techniques. We trained the models on two data sets : one with detailed questionnaire values and another with less granular data, such as averages.

	Réseau Bayésien	Arbres de décision	Random Forest	K-Neighbors	Régression Logistique	SVM
CHAINS - détaillés	0.0833	0	0.1667	0.0833	0	0
CHAINS - moyennes	0.0667	0	0	0.0667	0.1333	0
OVR - détaillés	0.0833	0	0.0833	0.0833	0	0
OVR - moyennes	0.0667	0	0	0.0667	0.1333	0.0667

Figure 2: Results of the machine learning models

The Random Forest model, trained using the classifier chains technique, initially achieved the best performance with a reliability score of 0.1667. However, after further analysis of the data, we removed some values, as they were found to hinder the model's performance. Following this adjustment, the model accuracy improved to 0.3636. Although this score is still not optimal, it can be attributed to the limited amount of training data available. We also developed a web application to allow users input their results and view the conclusions predicted by the model. Additionally, design recommendations and techniques were displayed within the application, based on the areas of improvement identified by the model.

This study explores a new approach to help interface developers interpret quantitative UX questionnaire results and identify areas needing improvement. The results show that using a machine learning model is feasible, although accuracy could be improved with a larger Attrakdiff answers dataset and the incorporation of additional models. Future work could improve this analysis by implementing an HTML scanning method to gain more detailed insights into interface elements like buttons, colors, fonts, and other design components.

References

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